AMITY UNIVERSITY MAHARASHTRA
AMITY SCHOOL OF ENGINEERING & TECHNOLOGY
B. Tech Computer Science & Engineering
ACADEMIC YEAR: 2025-26(ODD)

Course	**:**	B.Tech (CSE)
Course Code &Subject:		CSE 2301 & DCCN
Semester & Year	**:**	5th Sem / 3rd year
Division/Batch	**:**	A

ASSIGNMENT-04

1. Develop a design for congestion control and avoidance in TCP networks. Include queuing mechanisms and QoS Considerations.
2. Explain in detail the working of application layer protocols HTTP, FTP, and SNMP. Compare their functionalities and use cases.
3. Describe in detail the architecture and working of Domain Name System (DNS). Explain recursive and iterative resolution, caching.
4. Design and analyze the operation of distance vector and link state routing algorithms. Explain how routing tables are built and maintained.
5. Evaluate congestion avoidance mechanisms such as leaky bucket and token bucket algorithms. How do they ensure QoS?
6. Explain the architecture of the Domain Name System (DNS). Discuss its role in resolving names to IP addresses.
7. Analyze congestion control mechanisms used in TCP. How does TCP maintain reliability and flow control during heavy traffic?
8. Design an IP addressing scheme for a university with four campuses interconnected by routers. Allocate subnets for each department, explain classless addressing (CIDR), and illustrate routing between subnets using a sample routing table.
9. Explain with diagram the working of HTTP and SMTP protocols. Compare their roles in data communication.